

Kim Nguyen

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EDUCATION

10/2023 – Present <i>Germany</i>	Julius-Maximilians-Universität Würzburg <i>Master of Molecular and Computational Biology (GPA: 1.7)</i>
10/2017 – 04/2022 <i>HCMC, Vietnam</i>	International University, Vietnam National University <i>Bachelor of Science: Biotechnology (GPA: 1.8)</i>

RELEVANT RESEARCH PROJECTS

12/2025 – 06/2026 <i>Germany</i>	Theodor Boveri Institute, Julius-Maximilians-Universität Würzburg <i>Master's Thesis (Grade: 1.3)</i> • Title: <i>Characterisation of the RNA-Binding Behaviour of the RNA-Binding Protein MATRIN3 in Murine Cells.</i> • Developed and optimized computational workflow for transcriptome-wide analysis integrating multi-dataset HyperTRIBE, iCLIP and bulk RNA-seq datasets to identify reproducible and biologically meaningful targets of MATR3-RBP with high confidence. • Analyzed iCLIP crosslink and binding sites using PureCLIP and BindingSiteFinder (Bioconductor); Performed feature selection and statistical modelling (R), differential gene expression analysis DESeq2. • Found that ~20% target mRNAs of MATR3 are involved in sarcomere and contractile function, ECM organization, and cardiomyopathy-related processes. • Inspected sequencing quality and genomic tracks (BAM, BigWig, BED) using IGV.
03 – 09/2025 <i>Germany</i>	Theodor Boveri Institute, Julius-Maximilians-Universität Würzburg <i>Computational RNA Biology Intern</i> • Project: <i>Regulatory Impact of 3' Splice-Site Factors on Gene Expression.</i> • Investigated the effects of U2AFs and SF1 on gene expression and alternative splicing using ENCODE RNA-seq datasets from HepG2 and K562 cells. • Executed differential expression analysis (DESeq2), and alternative-splicing analysis (MAJIQ, VOILA); characterize regulatory relationships within the splice-site factor network via clustering and correlation analysis. • Inspected sequencing quality, and splicing sites using IGV.
10/2023 – 02/2024 <i>Germany</i>	Bioinformatics Department, Julius-Maximilians-Universität Würzburg <i>Systems Biology Research Practical</i> • Constructed and analyzed metabolic networks of <i>Candida albicans</i> using YANAsquare and METATOOL. • Curated pathway data from multiple databases (KEGG, Rhea, NCBI, UniProt) and literature. • Performed Elementary Flux Mode analysis and interpreted metabolic flux distributions across fungal metabolic networks.

ADDITIONAL RESEARCH EXPERIENCE

Apr. 2022 – Oct. 2022 <i>HCMC, Vietnam</i>	Cell Biotechnology Laboratory, International University <i>Bachelor's Thesis</i> • Optimized polyphenol extraction from <i>Perilla frutescens</i> using Response Surface Methodology (RSM).
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- Assessed antioxidant activity using Folin–Ciocalteu and FRAP assays and evaluated DNA-protection effects in *S. cerevisiae* and *E. coli* models.

Aug. 2021 – Dec. 2021
HCMC, Vietnam

The Applied Biotechnology Institute
Bioinformatics Intern

- Performed molecular identification of fungal isolates using **Sanger sequencing and ITS-based sequence analysis**.
- Quality-controlled and manually edited sequencing chromatograms using **ATGC**, followed by sequence alignment and **BLASTn analysis against NCBI GenBank**.
- Evaluated candidate species using **sequence identity, query coverage and reference-sequence quality** to support molecular species identification.
- Built and curated a reference database of approximately **200 fungal species**, integrating sequence identifications, GenBank accession information and literature references.

TECHNICAL SKILLS & RESEARCH INTERESTS

Programming: R, Python, Bash, HTML/CSS

Bioinformatics: Bioconductor, DESeq2, GenomicRanges, MAJIQ, PureCLIP, BindingSiteFinder, IGV, Nextflow

Tools: Git, Linux, HPC/Slurm (FastQC, MultiQC, Trimmomatic, STAR, HTSeq-count, featureCounts, SAMTools), Conda, venv, MS Office

Interests: Multi-omics, RNA Biology, Alternative Splicing, AI/ML-Driven Bioinformatics, Systems Biology

Languages: Vietnamese (native), English (Proficiency), German (B1)

EXTRACURRICULAR ACTIVITIES

2021
Vietnam

Vietnam National University Academic Newsletter
Research Assistant

- Analyzed and summarized **scientific publications** to extract key insights for non-specialist audiences.
- Translated clinical reports and biomedical research on **COVID-19 therapies and vaccines** into Vietnamese, improving accessibility and knowledge dissemination.
- Publication: *Treatments for COVID-19 in the World, Vaccine Efficacy Against SARS-CoV-2 Virus Variants*.

2019
Vietnam

Vietnam's Amazing Student Contest, RMIT University
Third Prize Winner

- Project: Anti-microplastic pollution campaign.